

Key Ingredients for EEG Foundation Models: What Works - and What Doesn't

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Abstract—Electroencephalography (EEG) signals present well-known challenges for machine learning, including strong inter-subject variability, heterogeneous acquisition protocols, and a scarcity of labeled data. Although these factors may appear unfavorable for the development of foundation models, recent studies demonstrate that it is possible to train robust EEG models when appropriate design choices are made. In this paper, we review the key architectural and methodological elements underlying modern EEG foundation models. We examine tokenization approaches, positional encoding schemes, self-supervised pretraining objectives, and the properties of large-scale pretraining datasets used in state-of-the-art systems. Our comparative analysis highlights a clear shift toward large-scale pretraining and identifies design choices that support generalization across diverse EEG tasks. At the same time, we show that increasing model scale does not automatically translate into superior generalization compared to smaller models trained from scratch for specific tasks. We conclude by outlining the remaining challenges, particularly the need for data-efficient and lightweight adaptation strategies in low-data settings, as well as the lack of fair and comprehensive evaluation benchmarks.

Index Terms—Electroencephalography (EEG), foundation models, transformers, self-supervised learning, brain-computer interfaces

I. INTRODUCTION

Electroencephalography (EEG) offers a non-invasive window into brain activity with high temporal resolution. EEG plays a crucial role in numerous neurotechnology and diagnostic applications where signals need to be annotated or classified. However, developing robust machine learning models for EEG analysis faces significant challenges: high inter-subject variability, heterogeneous recording protocols, limited labeled datasets, and diverse electrode montages. Traditional supervised learning approaches often fail to generalize across these variations, limiting their practical utility.

Foundation models (*i.e.*, deep learning models pretrained on large-scale unlabeled data and later adapted for specific tasks) have transformed natural language processing and computer vision [1], [2]. This paradigm is now expanding into biosignal analysis, where recent work demonstrates that EEG foundation models can learn transferable representations that generalize across individuals and heterogeneous recording conditions.

However, the question of whether these models can be efficiently adapted to new tasks with small sample size or post-training remains largely unresolved.

The past three years have seen a rapid proliferation of EEG foundation model architectures. Early approaches such as BIOT [3] employed linear transformers for joint pretraining across diverse biosignals, while EEG2Rep [4] introduced masked segment prediction in latent space. Neuro-GPT [5] adapted decoder-only architectures with autoregressive pre-training while LaBraM [6] was the first work to introduce neural token prediction. Recent innovations have pushed the boundaries further. NeuroLM [7] and ELASTIQ [8] explore the integration of EEG signals with text as input to large language models by treating neural activity as a type of language. Architecturally, CBraMod [9] introduces criss-cross attention with asymmetric positional encoding, while EEG-Mamba [10] leverage state space models to capture spatiotemporal dependencies efficiently. LUNA [11] addresses electrode layout heterogeneity through learned query mechanisms. Other notable contributions include mixture-of-experts architectures in NeurIPT [12], structured sparse attention in CSBrain [13], and unprecedented pretraining scale in REVE [14].

Despite this rapid progress, the field lacks a systematic comparison of the architectural choices and design principles underlying these models. This paper discusses the key ingredients that enable effective EEG foundation models and analyzes current limitations, thus providing guidance for future research and practical applications.

II. KEY INGREDIENTS OF EEG FOUNDATION MODELS

Foundation models distinguish themselves from traditional task-specific models through their training paradigm: rather than being trained from scratch for a particular application (*e.g.*, sleep staging, motor imagery classification, or seizure detection), they first learn rich representations from large-scale unlabeled data through self-supervised learning (SSL), and are subsequently adapted to downstream tasks through fine-tuning or linear probing (*i.e.*, final head only fine-tuning, while keeping the rest of the model frozen). SSL pretext tasks are designed to be as complex as possible, so that any other task could be seen as simpler and thus benefit from the learnt representations. Before discussing the common design

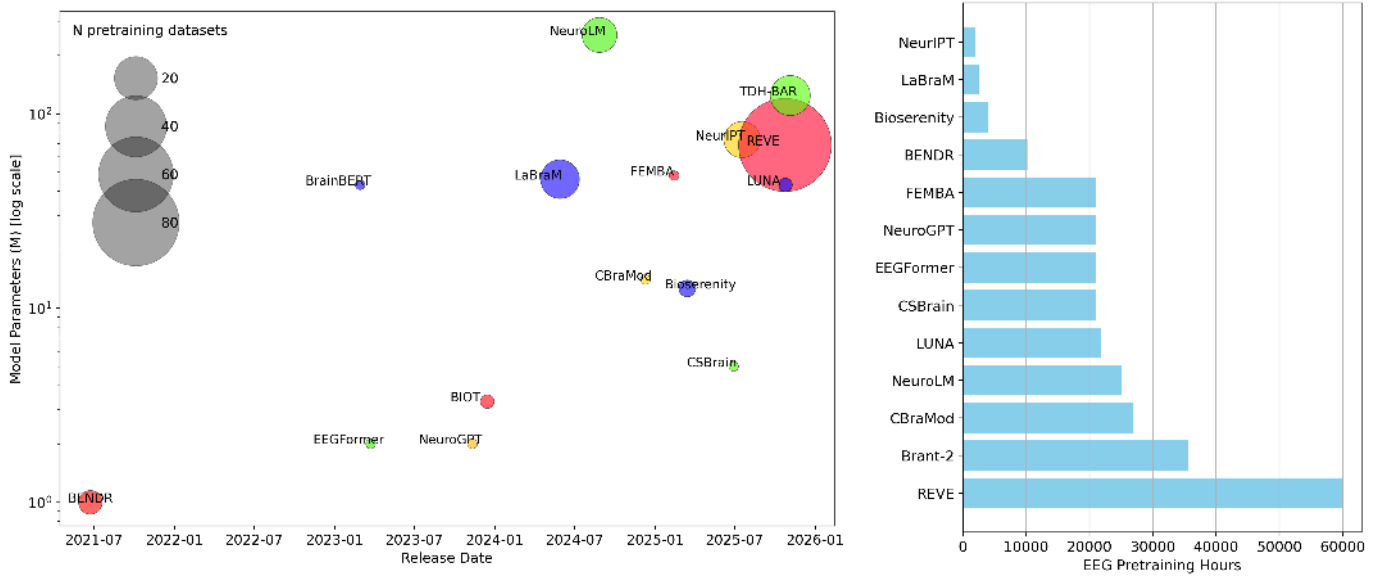


Fig. 1. EEG Foundation Models overview. Left panel: EEG Foundation Models plotted with respect to release date (x-axis) and model size (y-axis in Million of parameters). The bubble size indicates the number of of pretraining datasets. Right panel: Ranking with respect to the amount of pretraining hours.

choices shared by recently proposed EEG foundation models, it is important to highlight some key differences among them. A first distinction concerns the diversity of the pre-training data: some foundation models are trained on heterogeneous collection of datasets spanning multiple recording settings, whereas others are exposed to a single data source. A second, closely related aspect, is the ability to adapt to different downstream tasks, which in practice depends on how well a model can handle variability in EEG acquisition setups, most notably the number and arrangement of electrodes.

Several early foundation models, such as CBraMod [9] NeuroGPT [5] and EEGFormer [15], are pre-trained on a single large-scale dataset, the Temple University Hospital EEG Corpus (TUEG) [16]. Most of these approaches select a fixed subset of channels, which limits their ability to generalize to unseen configurations at inference time. More recent models, including LaBraM [6] REVE [14] and LUNA [11], adopt transformer-based architectures that relax the constraint of a fixed electrode layout and enable flexible handling of varying input configurations. These models are explicitly trained on data with varying input sizes, allowing them to operate on arbitrary channel sets at inference.

We identify five essential components that characterize EEG foundation models: (1) tokenization strategies for converting continuous multichannel signals into discrete tokens, (2) transformer architectures that process these tokens, (3) positional encoding for capturing spatiotemporal structure, and (4) SSL learning objectives that drive representation learning. We now examine each component in detail.

A. Tokenization

Tokenization transforms continuous multichannel EEG signals into discrete token sequences suitable for transformer processing. This crucial preprocessing step significantly im-

pacts model performance and computational efficiency. A straightforward strategy relies on fixed-size temporal patching, in which each EEG channel is divided into short temporal segments. Many approaches adopt this scheme, typically using relatively short windows (*e.g.*, 1 second), and treat the resulting spatiotemporal patches as continuous-valued tokens. More sophisticated methods learn neural tokenizers that adaptively compress EEG data. For instance, LaBraM [6], BrainOmni [17] and LUNA [11] employ vector quantization to create discrete codebooks. While this discretization introduces an explicit symbolic representation, the extent to which it provides a clear advantage over continuous spatiotemporal embeddings remains an open question, as competitive performance has also been reported using simpler continuous tokenization schemes.

B. Transformer Architectures

Transformer-based backbones are ubiquitous in current EEG foundation models. In most existing approaches, the Transformer serves as the core architecture, leveraging self-attention mechanisms to capture long-range dependencies in EEG time series. However, model sizes vary substantially, ranging from less than one to hundreds of million parameters (see Figure 1) and there are some architectural variations to address EEG-specific characteristics.

Most EEG foundation models adopt an encoder-decoder architecture during pretraining; after training, the decoder is discarded and only the pretrained encoder is retained for downstream tasks. In BIOT [3], linear transformers reduce the quadratic complexity of self-attention, enabling efficient processing of long EEG recordings. Decoder-only architectures that mirror autoregressive language models are used in NeuroGPT [5] while state space models such as Mamba [10]

offer an alternative to attention mechanisms, providing linear-time inference and strong performance on long sequences.

Specialized attention patterns have also emerged: criss-cross attention [9] separately models temporal and spatial relationships and structured sparse attention [13] exploits the hierarchical organization of brain activity. Mixture-of-experts architectures [12] route different input patterns to specialized subnetworks, potentially capturing the diverse nature of neural activity patterns involved in different tasks.

C. Positional Encoding

Earlier EEG deep learning models, primarily based on convolutional neural networks, were constrained in their ability to handle varying channel configurations due to their fixed architectural structure. In contrast, Transformer-based backbones operate on variable-length token sequences, enabling more flexible strategies for handling heterogeneous channel configurations. Among these, positional encoding, a standard component in state-of-the-art vision and language models, has emerged as a widely adopted approach also in EEG foundation models. Spatio-temporal positional encoding enriches EEG tokens with information about their temporal indices and spatial arrangement, allowing the model to capture both the temporal dynamics of EEG signals and the topographical structure of electrode layouts. This component is essential for effective EEG modeling, as meaningful patterns emerge from the joint interplay of temporal and spatial contexts.

Most existing EEG foundation models rely on sinusoidal positional encodings [11], [12] derived from Fourier features [1]. This choice offers several practical advantages for EEG analysis, as it naturally supports variable sequence lengths and channel counts while providing translation invariance in both temporal and spatial dimensions. Alternative approaches adopt learnable positional embeddings [9], which capture task-specific spatiotemporal regularities but tend to generalize less well to new tasks as empirical evidence suggests [12]. Hybrid strategies have also been explored: for example, REVE [14] combines sinusoidal and learned positional encodings. Additionally, asymmetric positional encoding schemes [9] explicitly differentiate between temporal and spatial dimensions, reflecting their fundamentally different properties in EEG data.

D. Self-Supervised Learning Objectives

In supervised learning, labeled data guide the model to focus mainly on features related to the target label, often ignoring the surrounding context. In contrast, SSL tasks that involve predicting missing parts of an image, reconstructing masked text, or understanding temporal relationships, encourage the model to learn richer, more complete representations. Among these, masked autoencoding [2] is by far the most widely adopted strategy: portions of the input are randomly masked, and the model is trained to reconstruct the missing content. Solutions that apply block-wise spatiotemporal masking rather than independent random masking [14], [18], together with masking ratios of approximately 50% [12], [14], have been shown to be particularly effective for EEG data. Across EEG

foundation models, a key point of divergence lies in how the input signal is masked and reconstructed. Raw signal reconstruction is the most prevalent strategy, employed by representative models such as Brant, CSBrain, EEGMamba, REVE. Other models rely on frequency-domain representations. For example, LaBraM formulates masked reconstruction in the spectral domain by predicting a learned neural codebook derived from the Fourier amplitude and phase, while Brant-2 [19] embeds power spectral density features of each projected raw-data patch. Similarly to raw signal reconstruction, token reconstruction (BENDR, BIOT, LUNA) operates in a learned embedding space (*i.e.*, the model learns to reconstruct the embedded representations). Other models such as LaBraM, BrainOmni, NeuroLM and THD-BAR use codebook index supervision where pretraining introduces discrete units that can be predicted as indices or reconstructed as codebook embeddings. Using a codebook is supposed to suppress low amplitude noise but it introduces additional hyperparameters and its added value in term of performances with respect to raw signal reconstruction is not demonstrated (see section III).

Learning objectives different from masked prediction have also been explored. Autoregressive prediction [5], for instance, trains models to predict future signal segments from past context, thereby explicitly modeling temporal dependencies. Unlike most EEG foundation models, EEGMamba [10] does not rely on SSL and shows that supervised learning across multiple datasets and electrode configurations can nevertheless achieve strong performance.

III. COMPARATIVE ANALYSIS

A. Pretraining datasets size and diversity

We first compare the scale and diversity of EEG data assembled for pretraining. Figure 1 illustrates the variety of datasets and amount of pretraining hours across recent models.

Early EEG foundation models, such as BENDR and LaBraM, were pre-trained on relatively limited amounts of data, typically fewer than 10,000 hours of recordings and involving a comparatively small number of subjects, although precise details regarding the size of the pre-training datasets are not always reported. More recent models have substantially increased both the scale and/or diversity of their training data. For example, LaBraM uses roughly 2,500 hours drawn from 16 datasets. REVE [14] leverages recordings from approximately 25,000 subjects, totaling around 60,000 hours of EEG data collected across 92 different datasets, with an unprecedented pretraining scale and diversity. If this progression reflects a growing recognition that the large variability of EEG signals requires extensive and diverse training data to achieve robust generalization, in practice larger pretraining do not necessarily yields better performances (see also Table I.) The impact of data scale and diversity in pre-training can vary depending on the downstream task [20]. In particular, subject diversity appears to be a key factor for effective scaling, while careful data curation may play an equally important role. This is demonstrated in [21], where training on only 5% of a 2,500-hours dataset, selected based on diversity of data and rejection

of outliers, achieves performance comparable and in some cases better than training on the full dataset.

B. Performances

Table I presents a performance comparison of recent EEG foundation models across nine diverse datasets spanning multiple applications: motor imagery classification [22], [23], sleep staging [24], epilepsy detection [16], emotion recognition [25], [26], mental disorder detection [26], stress assessment [27], and event-related potentials classification [28]. We maintain strict evaluation consistency by using data splits established in prior work. Our comparison includes non-foundation architectures such as EEGNet [29] and EEGConformer alongside recent foundation models. This table is based on results reported in [14] together with performance figures from more recent foundation models. Overall, the most recent models incorporating the main ingredients discussed above and pre-trained at scale tend to achieve the strongest performance on the selected downstream tasks. However, the table also highlights significant challenges in conducting fair comparisons between models. First they have not been evaluated on the same downstream tasks, as reflected by the large number of missing entries, and they are often fine-tuned using different protocols. In principle, foundation models are expected to reduce reliance on task-specific labeled data and to enable effective adaptation with minimal calibration. However, only very few works, such as REVE and CBraMod, report results under linear probing. This observation challenges the idea that EEG foundation models are readily reusable in low-data regimes and underscores the need for standardized evaluation benchmarks.

IV. OPEN CHALLENGES AND FUTURE DIRECTIONS

Despite recent progress, several important challenges remain for EEG foundation models. As clearly emerged in the tentative we made here to compare model performances, one of the most critical limitations is the *absence of a unified and standardized benchmark for evaluation*, comparable to meta-datasets used in computer vision. Such a benchmark would need to span multiple datasets while enforcing strict separation between pre-training data and downstream test sets. It is also necessary to define fine tuning and linear probing condition to equally quantify the resources (in terms of dataset size and number of training epochs) needed to adapt foundation models to new downstream tasks as it is especially relevant for real case deployment. This lack of standardization makes it particularly difficult to fairly compare recent foundation models. Furthermore given the large size of EEG foundation models and the limited availability of labeled EEG data, efficient adaptation techniques are crucial. Parameter-efficient fine-tuning methods such as LoRA [30] are promising directions, but systematic studies on how to best adapt EEG foundation models remain limited. Recent work such as NeuroTTT [31] represents a step in this direction, but further investigation is needed.

Secondly, it is important to underline that, despite their promise, *EEG foundation models do not yet consistently outperform task-specific models* trained directly on target datasets in cross-subject settings. Traditional machine learning methods still outperform foundation models on within subject setups and are sometimes on par with them, depending on the downstream tasks [32]. For example, on the BCI IV-2a dataset, a simple convolutional neural network EEG-SimpleConv [33] achieves a balanced accuracy of 0.72 compared to 0.65 for REVE, whereas on larger datasets such as PhysioNet-MI the two approaches yield comparable performance. An additional limitation arises from the evaluation setup: despite the versatility expected of foundation models, that should handle heterogeneous downstream tasks (segmentation, object detection, question answering, etc.) *existing EEG foundation models are assessed only on classification tasks*, which questions the extent to which they can be considered foundational.

V. CONCLUSION

This paper systematically reviews the key ingredients underlying modern EEG foundation models, analyzing tokenization strategies, positional encoding schemes, transformer architectures, self-supervised objectives, and pretraining datasets. Our comparative analysis reveals a clear evolution toward massive-scale pretraining on diverse datasets. The performance analysis reveals that while larger datasets generally improve generalization, architectural choices also play crucial roles. Models specifically designed for electrode layout flexibility (LUNA, REVE) excel in cross-dataset transfer, while those with sophisticated attention mechanisms (CBraMod, CSBrain) often achieve competitive performance with fewer parameters. The field has progressed remarkably in just three years, from early proof-of-concept models to sophisticated architectures handling unprecedented data heterogeneity. Nevertheless, significant challenges remain, particularly in improving fast task-specific adaptation in low-data regimes and in enabling fair and comprehensive cross-model evaluation. Addressing these issues will be critical to fully realizing the potential of foundation models for advancing both neuroscience research and clinical neurotechnology.

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TABLE I

COMPARISON OF BALANCED ACCURACY ACROSS 9 EEG CLASSIFICATION TASKS. FOUNDATION MODELS ARE IN BOTTOM PART OF THE TABLE. BEST PERFORMANCE PER DATASET IS HIGHLIGHTED IN BOLD. PRETRAINED FOUNDATION MODELS GENERALLY SHOW BETTER TRANSFER LEARNING CAPABILITIES IN CROSS SUBJECT EVALUATION SETTINGS.

Method	TUAB	TUEV	PhysioNetMI	BCI-IV-2a	FACED	ISRUC	Mumtaz	MAT	BCI-2020-3
EEGNet	0.7642	0.3876	0.5814	0.4482	0.4090	0.7154	0.9232	0.6770	0.4413
EEGConformer	0.7758	0.4074	0.6049	0.4696	0.4559	0.7400	0.9308	0.6805	0.4506
SPaRCNet	0.7896	0.4161	0.5932	0.4635	0.4673	0.7487	0.9316	0.6879	0.4426
ContraWR	0.7746	0.4384	0.5892	0.4678	0.4887	0.7402	0.9195	0.6631	0.4257
CNN-Transformer	0.7777	0.4087	0.6053	0.4600	0.4697	0.7363	0.9305	0.6779	0.4533
FFCL	0.7848	0.3979	0.5726	0.4470	0.4673	0.7277	0.9314	0.6798	0.4678
ST-Transformer	0.7966	0.3984	0.6035	0.4575	0.4810	0.7381	0.9135	0.6631	0.4126
BIOT	0.7959	0.5281	0.6153	0.4748	0.5118	0.7527	0.9358	0.6875	0.4920
LaBraM-Base	0.8140	0.6409	0.6173	0.4869	0.5273	0.7633	0.9409	0.6909	0.5060
CBraMod	0.8289	0.6671	0.6417	0.5138	0.5509	0.7865	0.9560	0.7256	0.5373
REVE-Base	0.8315	0.6759	0.6480	0.6396	0.5646	0.7819	0.9644	0.7660	0.5635
LUNA-Base	0.8196	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
CSBrain	0.8172	0.6903	0.6304	0.5657	0.5752	0.7925	0.9643	N/A	0.6004
NeurIPT	0.8293	0.6761	N/A	0.5504	N/A	N/A	0.9803	N/A	N/A

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